



GOVERNMENT POLYTECHNIC, AMRAVATI
(AUTONOMOUS INSTITUTE OF GOVERNMENT OF MAHARASHTRA)

CURRICULUM DEVELOPMENT CELL

**PROGRAMME TITLE: DIPLOMA IN COMPUTER ENGINEERING /
INFORMATION TECHNOLOGY**

COURSE CODE: FC5473

COURSE TITLE: WEB APPLICATION DEVELOPMENT USING PHP

TEACHING SCHEME:

LEVEL OF COURSE	PRERE- QUISITE	WEEKLY CONTACT HRS.			TOTAL CREDITS	TOTAL WEEKS	TOTAL CONTACT HOURS		
		L	T	P			L	T	P
V	FC3411	03	-	02	05	16	48	-	32

EXAMINATION SCHEME:

THEORY(Marks)					PRACTICAL(Marks)		TOTAL (Marks)
ESE PAPER HOURS	ESE		PA	TOTAL	ESE	PA	
3Hrs	MAX.	70	30*	100	25#	25^	150
	MIN.	28	---	40	10	10	

@: Internal Assessment #: External Assessment: Practical Based

(*) Under the Theory PA, Out Of 30 Marks, 20 Marks is the Average of Two Tests and 10 Marks are for Micro project-

(^) Under practical PA Continuous Assessment of Practical Work is to be done by Course Teacher as per CDC norms.

For the courses having only practical examination, PA has two parts (i) Continuous Assessment of Practical work - 60% and (ii) microproject-40%.

1.RATIONALE:

PHP is a general purpose, server-side scripting language run a web server that's designed to make dynamic pages and applications. PHP as a web development option is secure, fast and reliable. In the growing field of Web technology it is essential for every Diploma pass outs to learn PHP Language to help them build interactive web applications. This course is designed to inculcate web based applications development skills in students using server side scripting with PHP.

2. COURSE OUTCOMES (COs)

At the end of this course, student will be able to: -

1. Develop program using control statement.
2. Perform operations based on arrays and graphics.
3. Develop programs by applying various object oriented concepts.
4. Use form controls with validation to collect user's input.
5. Perform database operations in PHP.
6. Develop programs to perform various operations on files.

3.DETAILED CONTENTS: THEORY

Unit	Unit Outcomes (UOs) (In cognitive domain)	Topic and Sub-topics	CO No.	Hr	Marks
Unit 1 Expression and Control Statement in PHP	1a. Write simple PHP program to solve the given expression. 1b. Use relevant decision Making control Statement to solve the given problem 1c. Solve the given Iterative problem using relevant loop Statement.	1.1. History and Advantages of PHP, Syntax of PHP. 1.2. Variables, Data types, Expressions and operators, constants 1.3. Decision making Control statements if, if-else, nested if, switch, break and Continue Statement. 1.4 Loop control structure while, do-while, for and for each loop.	1	06	10
Unit 2 Arrays, Functions and Graphics	2a. Manipulate the given Type of arrays to get the desired result. 2b. Apply implode, Explode functions on the given array. 2c. Apply the given string functions on the character array. 2d. Scale the given image Using graphics concepts/ functions.	2.1. Creating and Manipulating Array, Types of Arrays- Indexed , Associative and Multi-dimensional arrays 2.2. Extracting data from arrays, implode, explode, and array flip. 2.3. Traversing Arrays 2.4. Function and its types User defined function, Variable function and Anonymous function. 2.5. Operations on String and String functions : str_word count(), strlen(), strrev(), strpos(), strreplace(), ucwords(), strtoupper(), strtolower(), stremp(). 2.6 Basic Graphics Concepts, Creating Images, Images with text, Scaling Images, Creation of PDF document. Unicode.	2	08	14

Unit	Unit Outcomes (UOs) (In cognitive domain)	Topic and Sub-topics	CO No.	Hr	Marks
Unit 3 Apply Object Oriented Concepts in PHP	3a. Write constructor and destructor functions for the given problem in PHP. 3b. Implement inheritance to extend the given base class. 3c. Use overloading / overriding to solve the given problem. 3d. Clone the given Object	3.1. Creating Classes and Objects 3.2. Constructor and Destructor 3.3. Inheritance, Overloading And Overriding, Cloning Object. 3.4. Introspection, Serialization	3	08	14
Unit 4 Creating and validating forms	4a. Use the relevant form controls to get user's input. 4b. Design web pages Using multiple Forms For the given problem. 4c. Apply the given Validation rules on form. 4d. Set/ modify/ delete cookies using cookies attributes. 4e. Manage the given session using session variables	4.1. Creating a webpage using GUI Components, Browser Role-GET and POST methods, Server Role 4.2. Form controls: text box, text area, radio button, check box, list, buttons 4.3. Working with multiple forms A web page having many forms A form having multiple submit buttons. 4.4. Web page validation. 4.5. Cookies - Use of cookies, Attributes of cookies, create cookies, modify cookies value, and delete cookies. 4.6. Session - Use of session, Start session, get session variables, destroy session. 4.7. Sending E-mail.	4	08	10
Unit 5 Database Operation	5a. Create database for the given problem using PHP script 5b. Insert data in the given database using PHP script. 5c. Apply the specified update operation in database record using PHP script. 5d. Delete the given record	5.1. Introduction to MySQL Create a database. 5.2. Connecting to a MySQL database: MySQL Database Server from PHP. 5.3. Database operation Insert data, Retrieving the Query result. 5.4. Update and delete operation on table data.	5	12	14

	from the database using PHP script.				
Unit 6 Files and Directories	6a. Use different modes on files. 6b. Perform operations such as reading opening, reading operation on file. 6c. Perform uploading and renaming of files and directories.	6.1. Files, different modes of files, File permission, Writing Files, File Open, File Read, File Close, Directories, 6.2. Handling uploads, Renaming & Deleting Files & Directories.	6	06	08

4. LIST OF PRACTICALS:

Sr No.	PRACTICAL OUTCOMES (PrOs)	CO NO.
1.	a. Install and configure PHP, web server, MYSQL b. Write a program to print "Welcome to PHP". c. Write a simple PHP program using expressions and operators.	1
2.	Write a PHP program to demonstrate the use of Decision making control structures using- a. If statement b. If-else statement c. Switch statement	1
3.	Write a PHP program to demonstrate the use of Looping structures using- a. While statement, b. Do-while statement c. For statement d. For each statement	1
4.	Write a PHP program for creating and manipulating- a. Indexed array b. Associative array c. Multidimensional array	2
5.	a. Write a PHP program to - i. Calculate length of string. ii. Count the number of words in string -without using string functions. b. Write a simple PHP program to demonstrate use of various Built-in string functions.	2
6.	Write a PHP program to- Inherit members of super class in subclass. Create constructor to initialize object of class by using object oriented concepts	3

Sr No.	PRACTICAL OUTCOMES (PrOs)	CO NO.
7.	Design a web page using following form controls: a. Text box b. Radio button c. Check box d. Buttons	4
8.	Write simple PHP program to - a. Set cookies and read it. b. Demonstrate session Management.	4
9.	Develop a simple application to- a. Enter data into database b. Retrieve and present data from database.	5
10.	Develop a simple application to Update, Delete table data from database.	5
11.	Write a PHP script to read the contents of the given file.	6
12.	Write a PHP script to write the contents of the given file.	6

Note

- The entire above listed practical's need to be performed compulsorily except star (*) marked, so that the students reach the 'Precision level of Dave's Psychomotor Domain Taxonomy.
- The Process and Product related skills associated with each practical outcome shall be assessed on basis of following performance indicators.

S. No.	Performance Indicators	Weightage in %
1.	Write appropriate code to generate desired output in web application	30
2.	Debug, Test and Execute the programs	30
3.	Presentation of Output	20
4.	Able to Answer to oral questions	10
5.	Submission of report in time	10
Total		100%

The above PrOs also comprise of the following social skills/attitudes which are Affective Domain Outcomes (ADOs) that are best developed through the laboratory/field based experiences:

- Handle HTML, PHP and MYSQL server.
- Experiments with web application using PHP.
- Plan, Develop, Validate and Host web application.
- Demonstrate working as a leader or a team member.
- Follow ethical practices.

The ADOs are not specific to any one PrO, but are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he undertakes a series of practical experiences over a period of time. Moreover, the level of achievement of the ADOs according to Krathwohl's 'Affective Domain Taxonomy' should gradually increase as planned below:

- 'Valuing Level' in 1st year
- 'Organising Level' in 2nd year and
- 'Characterising Level' in 3rd year

5. SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related co-curricular activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group and prepare reports of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- a. Prepare journal of practicals.
- b. Undertake micro-projects.

6. SUGGESTED INSTRUCTIONAL STRATEGIES

Following are suggested instructional strategies, which the teacher can adopt for the attainment of the various outcomes in this course:

- a. Massive open online courses (**MOOCs**) may be used to teach various topics/sub topics.
- b. About 10-15% of the topic/ subtopic/ content which is relatively simple and descriptive in nature may be given to the students for self learning, for self directed learning and assess the development of CO's through class room presentation.
- c. Guide student(s) in undertaking micro-projects.
- d. Demonstrate students thoroughly before they start doing the practice.
- e. Use different instructional strategies in classroom teaching.
- f. Observe continuously and monitor the performance of student in lab.

7. SUGGESTED MICRO-PROJECTS.

Only one micro-project is to be undertaken by a student in the beginning of the semester. S/he ought to submit it by the end of semester. Each micro-project should include two or more Cos. The micro-project could be industry application based, internet based, laboratory based, field based and survey based. In the all semesters, it could be group-based (5 to 6 Student) to build up skill and confidence in every student. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The student ought to submit micro-project by the end of the semester to develop the industry oriented COs. A suggested list is given below.

- a) Develop an application for student attendance management system.
- b) Develop an application for-
- c) sending plain text email.
- d) sending HTML message.
- e) sending emails with attachment
- f) Develop an application for Library Management system.
- g) Develop an application for Student feedback system.
- h) Develop an application for Hotel Management system.
- i) Develop an application for Online Auction system.
- j) Develop an application for Student Management system along with SMS based intimation.
- k) Develop an application for Training and Placement Management System for TPO Cell.
- l) Develop an application for Leave Management System.
- m) Develop an application for Local Search Engine.
- n) Develop an application for Library Management System.

8. MAJOR EQUIPMENTS/INSTRUMENTS REQUIRE

Sr No.	Equipment Name with Broad Specification	Practical No.
1	Hardware : Computer system (i3 - i5 preferable) (Any computer system with basic configuration)	All Practical
2	Operating system : Windows / Linux	
3	Any database tool such as MySQL, MariaDB or any equivalent tool	
4	Hardware : Computer system (i3 - i5 preferable) (Any computer system with basic configuration)	

9. SUGGESTED SPECIFICATION TABLE FOR QUESTION PAPER DESIGN

Unit No.	Unit Title	Marks per Unit	Distribution of Theory Marks			Total Marks
			R Level	U Level	A Level	
1	Expression and Control Statement in PHP	10	02	04	04	10
2	Arrays, Functions and Graphics And Graphics Functions and Graphics	14	06	04	04	14
3	Apply object oriented concepts in PHP Object Oriented Concepts in PHP	14	02	04	08	14
4	Creating and validating forms Creating and validating forms	10	02	04	04	10
5	Database Operations Operation	14	02	08	04	14
6	Files and Directories	08	02	02	04	08
Total		70	16	26	28	70

R= Remember, **U**= Understanding, **A**=Application and above (*Bloom's Revised taxonomy*)

Note: This specification table provides general guidelines to assist student for their learning and to teachers to teach and assess students with respect to attainment of UOs. The actual distribution of marks at different taxonomy levels (of R, U and A) in the question paper may vary from above table.

10. SUGGESTED LEARNING RESOURCES:

Sr. No.	Title of Book	Author	Publication
1.	Programming PHP	Rasmus Lerdorf, Kevin.T and Peter	O'Reilly, USA, ISBN -978-1-449 - 39277-2, 2018
2.	The Complete Reference PHP (Third Edition covers PHP)	M. Holzner, Steven	McGraw hill, New Delhi, ISBN 9780070223622, 2019.
3.	PHP and MySQL	McGrath, Mike	McGraw Hill, New Delhi, ISBN- 13: 978-1259029431
4.	Advance Web Technology	Dr. Rajendra Kawle	Devraj Publication , ISBN- 978-93- 86492-01-2

11. SOFTWARE/LEARNING WEBSITES.

- <http://spokentutorail.org>
- <https://www.w3schools.com/php/default.asp>

- c. <https://www.guru99.com/what-is-php-first-php-program.html>
- d. <https://www.tutorialspoint.com/php/>
- e. <https://tutorialehtml.com/en/php-tutorial-introduction/>
- f. www.tizag.com/phpT/
- g. <https://books.goalkicker.com/PHPBook/>
- h. <https://codecourse.com/watch/php-basics>

12. COURSE CURRICULUM DEVELOPMENT COMMITTEE:

SR. NO.	NAME	DESIGNATION	INDUSTRY/INSTITUTE
1	Mr. M. R. Torney	Lecturer in Computer Engineering	Government Polytechnic Amravati
2	Mr. R.R.Bhoge	Lecturer in Information Technology	Government Polytechnic Amravati

Govt. Polytechnic, Programme Board of Studies (Information Technology) has approved the above course curriculum on 14/05/2020 and is adopted for Information Technology Programme.

CHAIRMAN
PROGRAMME BOARD OF STUDIES,
COMPUTER ENGINEERING
GOVERNMENT POLYTECHNIC, AMRAVATI.

The General Board of Studies has approved the above course curriculum on 14/07/2020

The Governing Body has approved the above course curriculum on 13/08/2021